

PHISHING EMAIL DETECTION & AWARENESS REPORT

Cyber Security Task 2 – Future Interns

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Assessment Type: Phishing Email Detection &
Awareness System

1.Executive Summary

This report presents a structured analysis of phishing email samples collected from real-world scenarios. The objective is to identify phishing indicators, analyze email headers, classify risk levels, and provide awareness guidelines.

A total of 5 phishing email samples were analyzed, all demonstrating common attack techniques such as spoofed domains, urgency-based manipulation, and credential harvesting attempts.

Email header analysis confirmed authentication failures including SPF, DKIM, and DMARC misconfigurations, indicating high probability of email spoofing.

The findings highlight the importance of user awareness and email authentication mechanisms in preventing phishing attacks.

2.Introduction

Phishing is one of the most common and financially damaging cyber threats affecting individuals, businesses, and government organizations worldwide. Cybercriminals use deceptive emails, fake websites, malicious attachments, and social engineering techniques to trick users into revealing sensitive information such as passwords, banking details, credit card information, and One-Time Passwords (OTPs).

Unlike many cyberattacks that exploit software vulnerabilities, phishing primarily targets human behavior and trust. Attackers often impersonate legitimate organizations and create a sense of urgency, fear, or reward to manipulate victims into taking immediate action.

This report focuses on identifying phishing indicators, analyzing email samples, examining email header information, classifying email risks, and providing practical recommendations that help users recognize and avoid phishing attacks.

3.Aim of the task

The purpose of this task is to perform a phishing email assessment by analyzing suspicious email samples and identifying characteristics commonly used in phishing campaigns.

The assessment aims to:

- Study real-world phishing email examples
- Identify common phishing indicators
- Analyze email header information
- Classify emails based on risk level
- Explain phishing techniques in simple and understandable language
- Promote safe email handling practices

4.Objectives

The primary objectives of this assessment are:

1. Collect and review phishing email samples.
2. Inspect sender information, links, and email content.
3. Analyze email headers using professional analysis tools.
4. Identify signs of phishing and social engineering.
5. Classify emails as Safe, Suspicious, or Phishing.
6. Document findings in a professional report.
7. Provide awareness and prevention recommendations.

5.Scope of Analysis

The scope of this assessment includes:

- Analysis of phishing email content
- Detection of suspicious sender domains
- Identification of malicious or deceptive links
- Review of social engineering techniques
- Email header investigation
- Risk classification of analyzed emails
- User awareness and prevention guidance

This assessment is conducted strictly for educational and cybersecurity awareness purposes.

6.TOOLS AND RESOURCES USED

Email Analysis Tools

- Google Message Header Analyzer
- MXToolbox Email Header Analyzer
- Browser-based URL Inspection Techniques

Sample Sources

- Public phishing email datasets
- Educational phishing email repositories
- Real phishing email header sample

Documentation Tools

- Microsoft Word
- PDF Export
- GitHub Repository

7.Methodology

The following methodology was followed during the assessment:

Step 1: Collect phishing email samples.

Step 2: Review email content and sender information.

Step 3: Analyze suspicious domains and URLs.

Step 4: Examine email header information.

Step 5: Identify phishing indicators.

Step 6: Classify risk levels.

Step 7: Document findings and awareness recommendations.

Step 8: Prepare final report and supporting evidence.

This methodology follows common practices used by security analysts when investigating email-based threats.

8.Phishing Email Sample Analysis

Sample 1 – Microsoft 365 Storage Quota Scam

From: support@microsoft-storage-alert.com

Subject: Mailbox Storage Limit Reached – Immediate Action Required

Dear User,

Your Microsoft 365 mailbox has exceeded its storage limit and incoming emails are being blocked.

To restore full access, please upgrade your mailbox storage immediately by clicking the link below: <http://microsoft-storage-upgrade.com>

Failure to complete this process within 24 hours may result in permanent email suspension.

Thank you,

Microsoft Support Team

Why This Is Malicious

- Sender domain is not an official Microsoft domain
- Creates urgency to force immediate action
- Generic greeting ("Dear User")
- Suspicious external URL
- Attempts to steal login credentials

Risk Level: HIGH (Credential Phishing)

Why High Risk?

- Targets Microsoft 365 login credentials
- Can provide attackers access to emails and corporate accounts
- May lead to identity theft and unauthorized account access
- Uses urgency to pressure users into clicking malicious links

Sample 2 – Courier Delivery Failure Scam

From: delivery@express-courier-update.com

Subject: Delivery Attempt Failed – Confirm Your Address

Dear Customer,

We attempted to deliver your package today but were unable to complete the delivery due to an address verification issue.

Please confirm your delivery information using the link below:

<http://express-courier-update.com/confirm>

Failure to verify your address within 12 hours may result in package cancellation.

Regards,

Delivery Support Team

Why This Is Malicious

- Fake courier company domain
- Requests personal information
- Creates urgency

- Generic greeting
- Suspicious verification link

Risk Level: MEDIUM-HIGH (Information Theft Phishing)

Why Medium-High Risk?

- May collect personal information such as name, address, and phone number
- Can be used for identity theft or future phishing attacks
- Exploits curiosity and concern about package deliveries
- Less critical than banking credential theft but still dangerous

Sample 3 – Scholarship Award Scam

From: awards@international-scholarship-board.net

Subject: Congratulations! Scholarship Award Approved

Dear Student,

We are pleased to inform you that you have been selected to receive an international educational scholarship.

To receive your scholarship funds, complete your verification process immediately using the secure portal below:

<http://scholarship-award-verification.net>

Verification must be completed within 48 hours to avoid cancellation.

Congratulations once again.

Scholarship Processing Department

Why This Is Malicious

- Unexpected financial reward
- Unverified sender domain
- Requests personal information
- Creates urgency
- Suspicious website

Risk Level: HIGH (Financial Information Phishing)

Why High Risk?

- Targets students and job seekers who may be more likely to trust the offer
- May collect banking information and identity documents
- Uses a financial reward to manipulate victims
- Can lead to identity theft and financial fraud

Sample 4 – Corporate VPN Security Update Scam

From: it-support@vpn-security-update.com

Subject: Mandatory VPN Security Upgrade Required

Dear Employee,

As part of our cybersecurity enhancement initiative, all employees must update their VPN credentials immediately.

Please complete the upgrade process by visiting the secure portal below:

<http://vpn-security-update.com/login>

Failure to comply within 6 hours will result in loss of remote access privileges.

Regards,

IT Security Department

Why This Is Malicious

- Impersonates internal IT department
- Credential harvesting attempt
- Fake security update
- Suspicious domain
- Pressure to act quickly

Risk Level: HIGH (Corporate Credential Phishing)

Why High Risk?

- Targets employee usernames and passwords
- May provide attackers access to company systems
- Can result in data breaches and unauthorized network access
- Frequently used in corporate phishing campaigns

Sample 5 – Tax Refund Notification Scam

From: refunds@tax-refund-processing.net

Subject: Tax Refund Approved – Claim Your Payment

Dear Taxpayer,

After reviewing your recent tax records, we have determined that you are eligible for a refund of ₹24,750.

To receive your refund, please verify your banking details using the secure link below:

<http://tax-refund-processing.net/claim>

Failure to complete verification within 24 hours may result in cancellation of your refund.

Sincerely,

Tax Refund Processing Department

Why This Is Malicious

- Fake government-related sender
- Requests banking information
- Uses financial incentive
- Suspicious domain
- Urgency tactic

Risk Level: HIGH (Financial Phishing)

Why High Risk?

- Attempts to collect banking and personal information
- Uses the promise of a tax refund to gain trust
- Can result in financial loss and identity theft
- Victims may unknowingly disclose sensitive information to attackers

9.Final Risk Classification Summary:

Sample	Category	Attack Type	Risk Level	Impact
Microsoft Storage Scam	Phishing	Credential Theft	High	Account takeover
Courier Scam	Phishing	Information Theft	Medium-High	Identity theft
Scholarship Scam	Phishing	Financial Fraud	High	Banking/data theft
VPN Scam	Phishing	Corporate Attack	High	System breach
Tax Refund Scam	Phishing	Financial Fraud	High	Money loss

10.Common Phishing Indicators

Indicator	Description
Fake Sender Domain	Email appears to come from a trusted organization but uses a fraudulent domain
Generic Greeting	Uses "Dear User" or "Dear Customer" instead of the recipient's name
Urgency Tactics	Pressures users to act immediately
Suspicious Links	Links redirect users to fake websites
Credential Requests	Attempts to steal passwords or account information
Unexpected Attachments	Files that may contain malware

11.Working of a Phishing Attack

1. Attacker sends a deceptive email to the target
2. User clicks a malicious link or opens an attachment
3. User is redirected to a fake website or malware is executed
4. Credentials or sensitive data are captured by the attacker
5. Attacker gains unauthorized access or performs fraud

12.Security Best Practices

- Always verify sender identity before taking action
- Avoid clicking unknown or shortened links
- Check URL carefully before entering credentials
- Enable Multi-Factor Authentication (MFA)
- Report suspicious emails to IT/security teams
- Do not download untrusted attachments
- Keep antivirus and system software updated
- Safe Practices vs Risky Behavior

Safe Practices

- Verify email source before responding
- Hover over links to inspect destination URLs
- Use official websites instead of email links
- Enable MFA for all important accounts
- Report suspicious emails immediately

Risky Behavior

- Clicking links without verification
- Sharing passwords or OTPs via email
- Opening unknown attachments
- Trusting urgent or threatening messages
- Entering credentials on unverified websites

13. Email Header Analysis

This section includes technical analysis of email headers to identify:

- Sender authenticity
- IP address tracing
- Domain verification
- Routing path inspection

A phishing email header was analyzed using **Google Message Header Analyzer** and **MXToolbox** to verify sender authenticity, authentication status, and domain reputation.

The email impersonated a financial rewards service (Bradesco Livelô) and attempted to mislead users into believing that reward points were expiring.

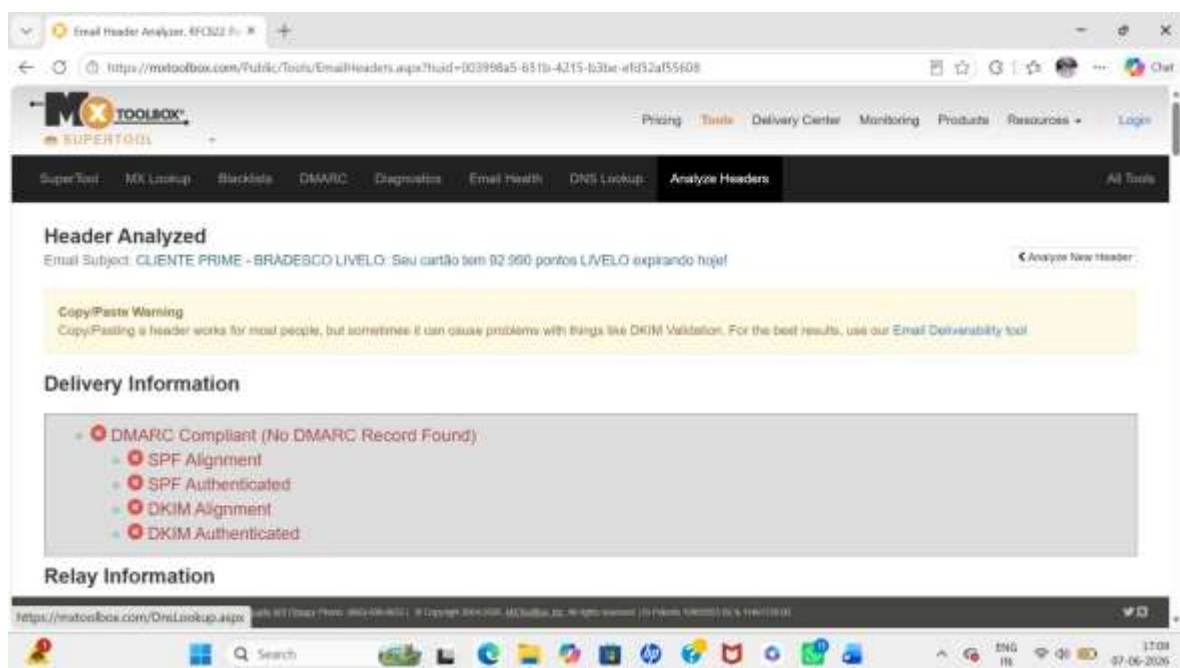
Header Analysis Summary

The following issues were identified during analysis:

- No valid DMARC record found for the sender domain
- SPF alignment issues detected
- DKIM alignment and authentication inconsistencies
- Sender domain is not properly configured for secure email delivery
- Missing or invalid DNS records for authentication checks
- The email appears to originate from a VPS-hosted or misconfigured server, commonly associated with spoofed or malicious email infrastructure.

These issues indicate that the email infrastructure is not legitimate and fails standard email security validation.

Figure 1: Google Message Header Analyzer Result

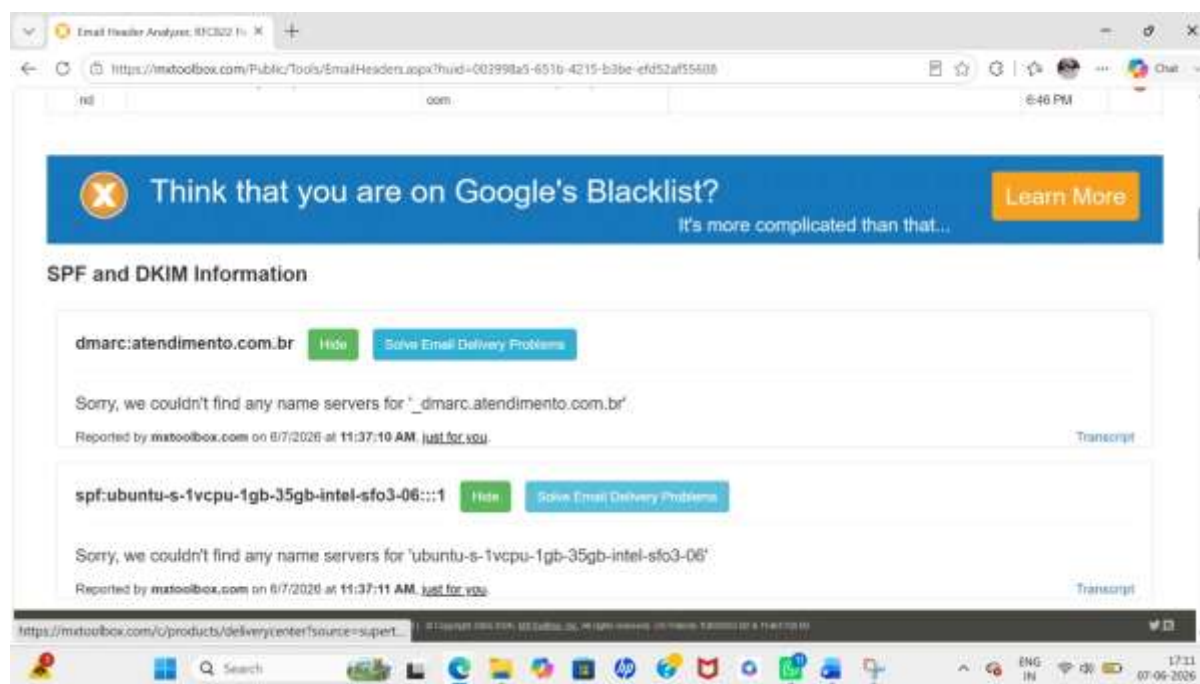


Observation:

The analysis shows multiple authentication problems including missing DMARC records, SPF alignment issues, and DKIM authentication warnings. Although some SPF/DKIM checks appear present, alignment failures indicate that the email is not properly authorized by the claimed domain.

These authentication failures strongly indicate domain spoofing and unauthorized use of infrastructure commonly associated with phishing campaigns.

Figure 2: MXToolbox SPF & DKIM Lookup Result



Observation:

The MXToolbox report shows missing or invalid DNS records for both DMARC and SPF verification. The system was unable to find proper nameserver records for the sender configuration, indicating weak or non-existent email security setup.

This is a strong indicator of a suspicious or malicious email source.

Risk Classification: HIGH RISK – Phishing Email

14.Conclusion

The analysis confirms that the email samples exhibit multiple phishing indicators including spoofed sender domains, weak or missing authentication mechanisms, and social engineering techniques such as urgency and reward-based manipulation.

Email header analysis further validates the presence of SPF, DKIM, and DMARC misconfigurations, indicating high likelihood of spoofed or malicious email infrastructure.

User awareness, verification of sender identity, and strict email authentication checks are critical to preventing such phishing attacks.